



KIGALI COLLEGE

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MACHINE LEARNING ASSIGNMENT PROJECT REPORT

1. Project Overview

This project develops a machine-learning model for predicting house prices in Rwanda. The target variable is House_Price_Million_RWF. The project uses multiple linear regression and includes a Streamlit web application for entering house information and receiving a predicted price.

2. Data Exploration and Cleaning

The dataset was examined and prepared before building the prediction model.

House_ID was excluded because it is an identifier rather than a useful predictive feature.

2.1 Cleaning Decisions

- The dataset was checked for extreme values that could affect the model.
- The data was reviewed for inconsistent values.
- The variables were prepared as regression-ready features.
- Categorical information such as Neighborhood was prepared for use by the regression model.
- The cleaned dataset was then used to build the prediction model.

In summary, the dataset was checked for extreme and inconsistent values. The data was cleaned, and the variables were prepared as regression-ready features before being used to build the model.

3. Model Building

A multiple linear regression model was used to predict house prices. The data was divided into training and testing sets using an 80/20 split with random_state=42.

- Area
- Bedrooms
- Bathrooms
- House Age
- Distance to City Centre
- Parking Spaces
- Neighborhood

4. Model Evaluation

The trained model was evaluated using the test dataset. The reported test performance was:

Metric	Result
R ²	0.7879
RMSE	22.94 million RWF

5. Saved Model and Streamlit Application

The trained model was saved as `house_price_model.sav`. The Streamlit application loads this saved model and provides input fields for the house characteristics. When the user clicks Predict, the application displays the predicted house price in million RWF.

6. Local Application Test

The application was tested locally using Streamlit. A prediction was successfully displayed in the browser.

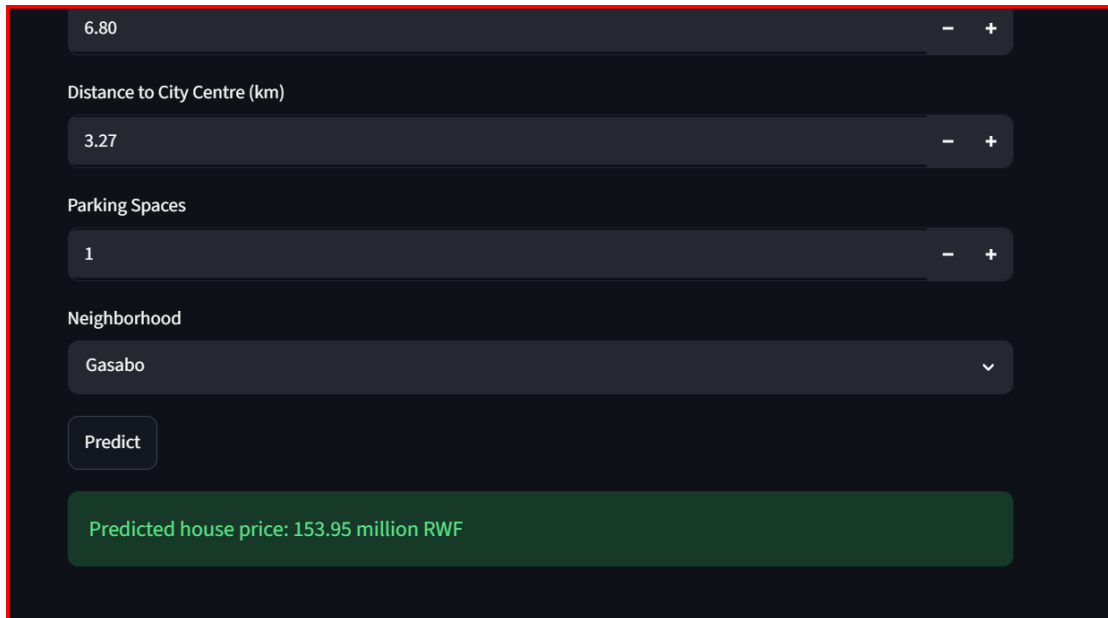
The image shows a Streamlit application interface with a dark theme. It features four input fields: a slider for 'Distance to City Centre (km)' set to 6.80, a slider for 'Distance to City Centre (km)' set to 3.27, a slider for 'Parking Spaces' set to 1, and a dropdown menu for 'Neighborhood' set to 'Gasabo'. Below these is a 'Predict' button. At the bottom, a green box displays the 'Predicted house price: 153.95 million RWF'.

Figure 1: Local Streamlit application showing a house-price prediction.

7. GitHub Repository and Deployment

The project files were uploaded to a public GitHub repository and deployed using Streamlit Community Cloud. The deployed application was tested online.

GitHub repository: <https://github.com/olidevoueux-oss/house-price-predictor>

Deployed application: <https://house-price-predictor-inet5odbec9gtwrffkjfps.streamlit.app/>

8. Online Application Test

The deployed application was opened online and tested with house information. The application successfully displayed a predicted price of 142.94 million RWF.

Figure 2: Deployed online Streamlit application showing a prediction.

9. Project Files

File	Purpose
app.py	Streamlit web application
house_price_model.sav	Saved trained machine-learning model
house_price_prediction_cleaned.csv	Cleaned dataset
Machine_Learning_Assignment_2_Solution.ipynb	Jupyter Notebook containing analysis and model development
requirements.txt	Python packages required to run the project
README.md	Project documentation and deployment information

10. Conclusion

The project successfully developed a Rwanda house-price prediction model using multiple linear regression. The dataset was cleaned and prepared for modelling, the model was evaluated using R^2 and RMSE, and the trained model was saved for use in a Streamlit application. The application was tested locally and deployed online.